

MTH 377/577 Convex Optimization

Problem set 1

Indicative solutions

1. The set

$$\{(a, b) \in \mathbb{R}^2 : b = 3a + 1\}$$

is **not** a vector space. Indeed, the zero vector $(0, 0)$ does not belong to the set since

$$0 \neq 3 \cdot 0 + 1 = 1.$$

Hence, the set does not satisfy the requirement of containing the zero vector.

2. The set

$$\{x \in \mathbb{R}^3 : \|x\| = 1\}$$

is **not** a subspace of $\mathbb{R}^3$. Indeed, the zero vector does not belong to this set since $\|0\| = 0 \neq 1$. Moreover, the set is not closed under scalar multiplication, as for any x with $\|x\| = 1$ and any scalar $\lambda \neq \pm 1$, we have

$$\|\lambda x\| = |\lambda| \neq 1.$$

3. (a) No.,

In equation form:

$$2x_1 + x_2 + 3x_3 + 2x_4 = 0$$

$$6x_1 - x_2 + 2x_3 + 5x_4 = 0$$

$$3x_1 + 4x_2 + x_3 + 4x_4 = 0$$

This system has a non-trivial solution. Hence, the vectors are linearly dependent. It can also be done by finding the RREF; if you get a pivot in every column, that would imply linear independence. Otherwise: Linear dependence.

- (b) Yes. Follow the same steps as part A.

4. There may be various answers. Compute the RREF, the non zero entries will form a basis for the span of the given vectors.
5. Yes. $C_1 = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 = 1\}$, $C_2 = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 = 100^2\}$. Consider $a \in C_1$, $b \in C_2$. Note that $d(a, b) \in \{99, 101\}$ for any a, b . The set of distances $\{99, 101\}$ is bounded.

[Additional explanation for understanding: Think of the two circles as concentric over the origin, 1 and 100 as the radii for each, Now the minimum distance you can be having between a point on the circle with radius 1 and radius 100 would be 99 if they lie on the same ray, and the max would be 101 if they would be lying on opposite rays, Hence it would be a bounded set $[99, 101]$.]

6. (a) The empty set $\emptyset$ is both **open** and **closed** in $\mathbb{R}$.
 (b) The complement of $\{x = 5\}$ in $\mathbb{R}$ is $\mathbb{R} \setminus \{5\}$, which is **open** (and not closed).

Remark: A set $S \subset \mathbb{R}$ is **open** (respectively, **closed**) if and only if its complement $\mathbb{R} \setminus S$ is **closed** (respectively, **open**) in $\mathbb{R}$.

7. In general, continuity at all points except one is not sufficient to guarantee the existence of a maxima or a minima. The Extreme Value Theorem (Bolzano-Weierstrass Theorem) requires the function to be continuous on the entire domain and the domain to be closed and bounded. See the **Fig.1**.

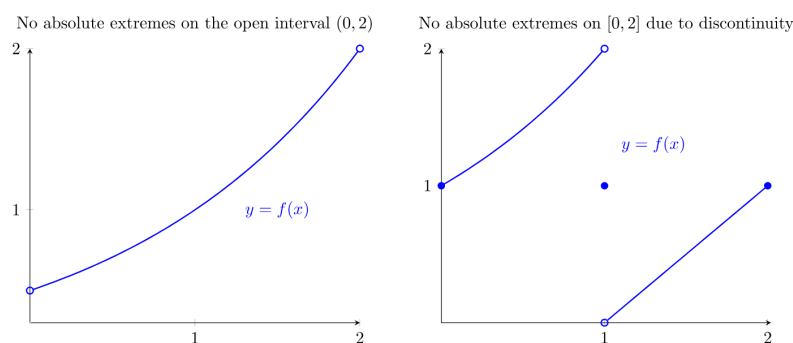


Figure 1:

8. Consider the set

$$A = \{x^k\}_{k=1,2,\dots} \subset \mathbb{R}, \quad \text{where } x^k = \sqrt{k}.$$

That is,

$$A = \{\sqrt{1}, \sqrt{2}, \sqrt{3}, \dots\}.$$

Recall that a set is closed in $\mathbb{R}$ if and only if every convergent sequence in the set converges to a point belonging to the set.

Let $\{y_n\}$ be a convergent sequence in A . Then there exists a sequence $\{k_n\} \subset \mathbb{N}$ such that

$$y_n = \sqrt{k_n} \quad \text{for all } n.$$

Since $\{y_n\}$ converges in $\mathbb{R}$, then $\{k_n\}$ must be bounded. Hence, $\{k_n\}$ is eventually constant, and therefore $\{y_n\}$ is eventually constant. In particular, its limit belongs to A .

Therefore, every convergent sequence in A converges to a point of A . Hence, A contains all of its limit points. Thus, A is a closed subset of $\mathbb{R}$.

9. **True.** Every affine set is convex. Indeed, if A is an affine set, then for any $x, y \in A$ and any $\lambda \in \mathbb{R}$,

$$\lambda x + (1 - \lambda)y \in A.$$

In particular, this holds for all $\lambda \in [0, 1]$, which is exactly the definition of convexity. Hence, every affine set is convex.

10. **Yes.** For any $x \in C_n$, $\sum_i ax_i \leq n$ for all $a \in I_+$. Note that this implies $C_n = \{(0, 0, \dots, 0)\}$: Let $x \in C$ such that some $x_i \neq 0$. Then for $\alpha > n$, $\sum_i ax_i > n$. Therefore $x \notin C$, we arrive at a contradiction. Similarly $C_{n+1} = \{(0, 0, \dots, 0) \in \mathbb{R}^{n+1}\}$. The complement of C_n and C_{n+1} are open sets since these are $\mathbb{R}^n \setminus \{0\}$ and $\mathbb{R}^{n+1} \setminus \{0\}$ respectively. Therefore, C_n and C_{n+1} are closed sets. [Alternatively, any convergent sequence in C_n, C_{n+1} must have all terms and limit as 0.]

Let A be a relevant domain where union of sets $\cup_n C_n$ is well defined. Union of closed sets is closed, therefore $C_n \cup C_{n+1}$ is closed. Let $m = 1$. $C_n \subset \{x \in \mathbb{R}^n | x_i \leq 1\}$. Therefore, C_n is bounded. Similarly, C_{n+1} is also bounded. Union of two bounded sets is bounded, therefore $C_n \cup C_{n+1}$ is bounded.

11. A, B are non-empty, disjoint and convex. Note that C is not closed. By the separating hyperplane theorems, we know that a weak separating hyperplane exists for A and B . Let $h = (1, 1)$, $\beta = 4$. (h, β) is the weak separating hyperplane. Note that this coincides with B .